ME2 — Vocabulary, Slots & Who Decides

The fixed 40-keyword vocabulary, the 19 intents — and exactly how much say Whisper gets

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The 40 fixed keywords

21 intent atoms point to an action · 19 slot values carry a parameter

play · music · next · pause · stop · volume · up · down · lights · on · off · call · message · reminder · timer · alarm · temperature · brightness · color · time · weather · 10 seconds · 30 seconds · 1 minute · 4 am · 8 am · 9 pm · 18 degrees · 22 degrees · 26 degrees · 20 percent · 60 percent · 100 percent · red · blue · green · yellow · drink water · study · exercise

Fixed at training time (string-matched against all 18 469 transcripts). The model can only hear words it knows — anything else is out-of-vocabulary.

What Whisper does

Audio → text caption (that's it)

- **faster-whisper base.en** (int8, ~74 M params) — encoder-decoder transformer, runs **on-device**, no cloud.
- After the wake word it writes the sentence: "*set alarm for 8 am*" — a subtitle, **not** a decision.
- **~1.1 s per clip** (slowest step) — excluded from every benchmark number.
- **Not an LLM**: no knowledge, no reasoning — it copies what it heard.

Where the caption is used

- **Display** — show "you said: …" in the log/GUI.
- **Free-form numbers** — slot policy `text`: "dim to **35** percent" (values outside the fixed 19) can override the keyword's value.
- **Second opinion on intent** — see the voting rules below.

Slots — 6 of the 19 intents take one

slot intent	values it accepts
TIMER	10 seconds · 30 seconds · 1 minute
ALARM	4 am · 8 am · 9 pm
TEMPERATURE	18 · 22 · 26 degrees

Whisper's voting power on the intent

1 · VCM (always runs) mel 40×250 → keyword sigmoids → rule → → **2 · Whisper (if enabled)** caption → text_intent() → **t_intent** → **3 · Guarded vote** see the 3 conditions → final intent

BRIGHTNESS	20 · 60 · 100 percent
COLOR	red · blue · green · yellow
CREATE_REMINDER	drink water · study · exercise

Other 13 intents (CALL, MESSAGE, PAUSE, STOP …) → slot **NONE**. No value keyword fired → NONE as well.

The 19 intents (output classes)

intent	slot	intent	slot_
PLAY_MUSIC	–	TIMER slot	yes
PAUSE	–	ALARM slot	yes
STOP	–	TEMPERATURE slot	yes
NEXT	–	BRIGHTNESS slot	yes
VOLUME_UP	–	COLOR slot	yes
VOLUME_DOWN	–	CREATE_REMINDER slot	yes
LIGHT_ON	–	TIME	–
LIGHT_OFF	–	WEATHER	–
CALL	–	LIST_REMINDERS	–
MESSAGE	–		

CALL & MESSAGE are action-only: the corpus has no caller/recipient names in the vocabulary.

ac_intent
+ ac_slot

Whisper's caption…	who wins
VCM is **unsure** (low confidence / no keywords)	**Whisper wins** — it rescues the utterance
caption is **strong** (clear command cue at the opening)	**Whisper wins**
caption **agrees** with the keyword head (expected keywords also fired)	**Whisper wins**
lone mishear, VCM confident, no agreement	**VCM wins — caption ignored**
caption says *CALL*, acoustics say colour (classic mishear)	**VCM wins** — CALL is exempt from the cue rule

In plain terms

- Whisper is a **tie-breaker / helper**, never a dictator: a wrong caption **cannot** overrule a confident keyword hit.
- Slots: with default `slot_policy=text` the caption's number may override the keyword's value — keywords still decide *which* slot.
- **`python app.py --mode mic --no-asr`** → Whisper never loads: intent + slots decided **100 % by the VCM** (the configuration for requirements #7/#8).

Benchmarks & poster accuracy (99.29 % / 99.46 %) are computed acoustic-only — Whisper is not in any reported metric. Requirements: #7 on-device ✓ · #8 pure VCM ✓ (with --no-asr) · #9 wake word allowed ✓